Minibatch Heavy-Ball SGD at Finite Dimension: Exact Threshold, Rate and Critical Slowing Down

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November 2026

Abstract

For heavy-ball SGD on a quadratic in $\mathbb{R}^P$ with minibatches of B random features having independent coordinates in the Hessian eigenbasis, coordinate-dependent fourth moments and a common random scale, we give at finite P and B the mean-square threshold via a rank-one secular equation and the exact convergence rate $\rho = 1/z_\star$, where $z_\star$ is the smallest positive root of $z\widehat{K}(z) = 1$ for the kernel $\widehat{K}$ of an exact convolution equation for the loss, together with the prefactor of the approach to the noise floor and the slope of $1-\rho$ at the threshold. Applied to the vocabulary model of Everett and Paquette [2026], where token rows are gated features sharing (η, β, B), this settles at every finite batch the three vocabulary statements that their Appendix A.4 lists as not proved, and the numerical observations of their Remark H.17: when the two extreme-frequency rows are stable, the slowest row is one of them; no (η, β) makes a gated feature decay faster per step than the best SGD step (proved at leading order there), so the fewest samples per decay time with momentum equal those of SGD; at fixed β the sample cost increases strictly with B, giving E_β in closed form; the tuned β is zero with $S_{\mathrm{mom}}(B) = S_{\mathrm{SGD}}(B)$ for $B - 1 \le B_\times$ (their batch size $B_\times$), and positive for $B - 1 > B_\times$; and the fewest steps approach their large-batch limit as $C_{\mathrm{mom}} B^{-1/2}$ with explicit C_{mom}. At high effective rank $r_{\mathrm{eff}} = \operatorname{tr} H/\lambda_1$ the threshold of the EMA step $\eta/(1-\beta)$ is $2B/(\gamma \operatorname{tr} H)$, independent of β and of the coordinate fourth moments, up to an explicit relative error that is $O(1/r_{\mathrm{eff}})$ when the coordinate kurtoses and $B(1-\beta)/(1+\beta)$ stay bounded.

1 Setting and result

Heavy-ball SGD on a quadratic with mean Hessian $H = \sum_{k=1}^P \lambda_k u_k u_k^\top$, $\lambda_1 \ge \cdots \ge \lambda_P > 0$, reads

$$w_{t+1} = \big((1+\beta)I - \eta H_{B_t}\big) w_t - \beta w_{t-1} - \eta \xi_t, \qquad \beta \in [0,1),\ \eta > 0, \tag{1}$$

with (H_{B_t}, ξ_t) i.i.d. across steps (arbitrarily coupled within a step) and independent of (w_0, w_{-1}), $\mathbb{E} H_{B_t} = H$, $\mathbb{E}\xi_t = 0$, $\Sigma_\xi = \mathbb{E}\xi_t \xi_t^\top \succ 0$. With $F_t = \left(\begin{smallmatrix} (1+\beta)I - \eta H_{B_t} & -\beta I \\ I & 0 \end{smallmatrix}\right)$, the second moment of $s_t = (w_t, w_{t-1})$ evolves through $Q \mapsto \mathcal{T}(Q) + \mathrm{blkdiag}(\eta^2 \Sigma_\xi, 0)$, up to a term vanishing with $\mathbb{E} s_t$, where $\mathcal{T}(Q) = \mathbb{E}[F_t Q F_t^\top]$; for $\Sigma_\xi \succ 0$ the second moments from $w_0 = w_{-1} = 0$ stay bounded if and only if $\rho(\mathcal{T}) < 1$, and for $\rho(\mathcal{T}) < 1$ they stay bounded from every initial condition, forcing correlated with H_{B_t} included (Appendix A; cf. Costa et al., 2005, Ch. 3 and Cai et al., 2026, Proposition C.3). In EMA form $m_{t+1} = \beta m_t + (1-\beta)(H_{B_t} w_t + \xi_t)$, $w_{t+1} = w_t - \eta_{\mathrm{EMA}} m_{t+1}$ reproduces (1) with $\eta_{\mathrm{EMA}} = \eta/(1-\beta)$.

Features. $H_{B_t} = \frac{1}{B}\sum_{b=1}^B x_{t,b} x_{t,b}^\top$ with i.i.d. $x = \zeta H^{1/2} \omega$, where the scalar ζ is independent of ω, $\mathbb{E}\zeta^2 = 1$, $\gamma = \mathbb{E}\zeta^4$, and the coordinates $\omega_k = u_k^\top \omega$ are independent with mean 0, variance 1 and $\mathbb{E}\omega_k^4 = 3 + \kappa_k$. This covers Gaussian ($\kappa_k = 0$) and Rademacher ($\kappa_k = -2$) features, per-coordinate

Bernoulli gates (large κ_k on rare coordinates), the per-sample gate of Everett and Paquette [2026] on an arbitrary H ($\gamma = 1/q$), and Student-t features. Put $p = 1/B$, $\kappa_\beta = \frac{1+\beta}{1-\beta}$, and

$$c_k = p\big(\gamma(2+\kappa_k)-1\big)\lambda_k^2, \quad \alpha_k = 1+\kappa_\beta\frac{c_k}{\lambda_k^2}, \quad a_k = 1+\beta-\eta\lambda_k, \quad S(\eta) = \frac{\eta p\gamma}{2(1-\beta)}\sum_{k=1}^{P}\frac{\lambda_k}{1-\frac{\eta\alpha_k\lambda_k}{2(1+\beta)}}.$$

Let χ_k be the mode-k impulse response of deterministic heavy ball ($\chi_k(0)=1$, $\chi_k(1)=a_k$, $\chi_k(\tau+1) = a_k\chi_k(\tau)-\beta\chi_k(\tau-1)$), $\psi_k = \eta^2\chi_k^2$, $h_k(\tau) = \psi_k(\tau)+c_k\sum_{s<\tau}\psi_k(\tau-1-s)h_k(s)$ and $K_\tau = p\gamma\sum_k\lambda_k^2 h_k(\tau)$. Their generating functions are rational:

$$\widehat{\psi}_k(z) = \frac{\eta^2(1+\beta z)}{(1-\beta z)\big(1-(a_k^2-2\beta)z+\beta^2z^2\big)}, \qquad \widehat{K}(z) = p\gamma\sum_k\frac{\lambda_k^2\,\widehat{\psi}_k(z)}{1-c_kz\widehat{\psi}_k(z)}.$$

Theorem 1. *Let $\sigma_k^2 = u_k^\top\Sigma_\xi u_k$, $\ell_t = \mathrm{tr}(H\mathbb{E}w_tw_t^\top)$ (twice the expected excess loss), and $\widetilde{\mathcal{M}} = \mathrm{diag}(\alpha_k\lambda_k) + p\kappa_\beta\gamma\sqrt{\lambda}\sqrt{\lambda}^\top$ with $(\sqrt{\lambda})_k = \sqrt{\lambda_k}$.*

(a) Threshold. *$\rho(\mathcal{T}) < 1$ if and only if $\eta < \eta^\star = 2(1+\beta)/\lambda_{\max}(\widetilde{\mathcal{M}})$, the unique root of $S(\eta) = 1$ in $\{\eta > 0 : \eta\,\max_k(\alpha_k\lambda_k) < 2(1+\beta)\}$. For $\eta < \eta^\star$,*

$$\ell_t \to \ell_\infty = \frac{f_\infty}{1-S(\eta)}, \qquad f_\infty = \frac{\eta}{2(1-\beta)}\sum_k\frac{\sigma_k^2}{1-\frac{\eta\alpha_k\lambda_k}{2(1+\beta)}}.$$

(b) Convolution equation. *From $w_0 = w_{-1} = 0$, the loss satisfies $\ell_t = f_t + \sum_{s<t}K_{t-1-s}\ell_s$ with $f_t = \sum_{\tau<t}\sum_k\lambda_k\sigma_k^2h_k(\tau)$; if every $c_k \geq 0$, then $K_\tau \geq 0$ and this is a renewal equation.*

(c) Rate. *If $\rho(\mathcal{T}) > 0$ (automatic unless $P = 1$, $\beta = 0$, $\gamma(3+\kappa_1) = 1$ and $\eta = 1/\lambda_1$), then $\rho(\mathcal{T}) = 1/z_\star$, with $z_\star$ the smallest positive root of $z\widehat{K}(z) = 1$.*

(d) Prefactor. *If $c_k \geq 0$ for all k ($\kappa_k \geq 1/\gamma - 2$, e.g. $\kappa_k \geq -1$) and $\eta < \eta^\star$, then $z_\star > 1$ is a simple root and the only one in $|z| \leq z_\star$, and from $w_0 = w_{-1} = 0$, with $\widehat{\phi}(z) = \sum_k\lambda_k\sigma_k^2\widehat{h}_k(z)$, $\widehat{h}_k = \widehat{\psi}_k/(1-c_kz\widehat{\psi}_k)$,*

$$\ell_\infty - \ell_t = C\rho(\mathcal{T})^t + O(R^{-t}), \quad R > z_\star, \qquad C = \frac{\widehat{\phi}(z_\star)}{(z_\star-1)\,\frac{d}{dz}\big(z\widehat{K}(z)\big)\big|_{z_\star}} > 0.$$

(e) Critical slowing down. *Under the condition of (d), as $\eta \uparrow \eta^\star$ (with K_τ at $\eta^\star$, where $\sum_\tau K_\tau = 1$ and the denominator is the mean renewal time),*

$$1-\rho(\mathcal{T}) = \frac{S'(\eta^\star)}{\sum_{\tau\geq0}(\tau+1)K_\tau}\,(\eta^\star-\eta) + O\big((\eta^\star-\eta)^2\big).$$

The key step is that $\mathcal{T}$ preserves the matrices with H-diagonal $P \times P$ blocks and has the same spectral radius there at every η; on that subspace it is block diagonal plus rank one (Lemma 8). Without the rank-one coupling each mode would be an independent scalar chain, as in the decoupled vocabulary of Everett and Paquette [2026, Lemma H.12]; with it, (c) follows from the convolution equation and (d)–(e) from its renewal form. For Rademacher modes ($c_k < 0$) K_τ can be negative (at $P = B = 1$, $\lambda_1 = 1$, $\gamma = 1$, $\kappa_1 = -2$, $\beta = 0$, $\eta = 0.8$: $\rho(\mathcal{T}) = 0.04$ and $K_1 = -0.384$), (a)–(c) still hold, and the approach to the floor need not be a single exponential: for $P = 1$ the iteration is deterministic and all restricted eigenvalues have modulus β in the underdamped regime.

2 A vocabulary: the slowest row is an extreme row

In the vocabulary model of Everett and Paquette [2026, Section H.3] a sample draws a token i with $\Pr(i = j) = q_j$ (their p_j), a Gaussian input $\tilde{x} \in \mathbb{R}^d$ and label noise, and only row i of the parameter matrix receives a gradient, with (η, β) shared across rows. The rows are coupled, but by their Lemma H.12 row j has the marginal law of the gated feature above with $H = q_j I_d$, $\gamma = 1/q_j$ and $\kappa_k = 0$ ($c_k = (2q_j - q_j^2)/B \geq 0$), and the loss decays at rate $\Lambda_{\min} = \min_j \Lambda_j$, where $\Lambda_j = -\ln \rho(T_j)$ for the linear part T_j of the row-j second-moment recursion. On the traced moments $\mathcal{T}$ restricts to

$$M(q) = \begin{pmatrix} a^2 + \eta^2 q(d+2-q)/B & -2a\beta & \beta^2 \\ a & -\beta & 0 \\ 1 & 0 & 0 \end{pmatrix}, \qquad a = 1 + \beta - \eta q,$$

with $\eta = (1-\beta)\eta_{\mathrm{EMA}}$; T_j is similar to $M(q_j)$, so $\Lambda_j = -\ln \rho(q_j)$ with $\rho(q) = \rho(M(q))$. Everett and Paquette [2026, Appendix A.4] write: "With momentum at a finite batch, that the slowest row is one of the two extremes, and how close $S_{\mathrm{mom}}(B)$ comes to its large-batch limit, are observed numerically." The results below concern the exact second-moment decay rate of this least-squares vocabulary model; they do not address the logistic-regression model of Everett and Paquette [2026] or the samples needed to reach a prescribed loss with label noise.

Everett and Paquette [2026]	This note
A.4: slowest row with momentum at a finite batch (observed numerically)	Theorem 2
A.4: critical batch size at fixed β (open, E_β not determined)	Corollary 4
A.4: curve of steps against batch size for momentum (not described)	Theorem 6(a)
A.4: how close $S_{\mathrm{mom}}(B)$ comes to S_{mom} (observed numerically)	Theorem 6(c)
H.17: $E_{\mathrm{mom}} = E_{\mathrm{SGD}}$; tuned β zero below and positive above $B - 1 = B_\times$	Corollaries 4, 5, Theorem 6(b)

Theorem 2. *Let $d \geq 1$, $B \geq 1$, $\eta > 0$ and $\beta \in [0, 1)$. For every $s < 1$ the set $\{q \in (0, 1] : \rho(q) \leq s\}$ is an interval. Consequently, if $q_{\min} \leq q_j \leq q_{\max}$ for every j and the rows of frequency $q_{\min}$ and $q_{\max}$ are mean-square stable, then every row is stable and $\max_j \rho(q_j) = \max\big(\rho(q_{\min}), \rho(q_{\max})\big)$: the slowest row is the most or the least frequent one.*

The theorem holds for every d, B, η and β, finite-batch gradient noise included; label noise does not enter T_j. Stability of the two extreme rows cannot be dropped: for $d = B = 1$, $\eta = 0.36878$, $\beta = 0.51037$, $\rho = 1.000195$, 1.016659, 0.932660 at $q = 0.001$, 0.2124, 1. That the stable frequencies form an interval is in Everett and Paquette [2026, Lemma H.13(i)], for SGD the theorem follows from their Lemma H.13(v), and without gradient noise the deterministic factor $\rho(\bar{F})$ decreases in curvature up to $1+\beta$ and increases after it [Everett and Paquette, 2026, proof of Proposition H.15]. In the proof (Lemma 9), $\rho(q) \leq 1/z$ reduces on $(1, 1/\beta)$ to $g_q(1/\eta, z) \geq 0$ for a quadratic in q negative at 0. Over 2163 random stable settings ($d \leq 200$, $B \leq 256$, $10^{-3} \leq \eta \leq 10$, $q \in [10^{-4}, 1]$) $\rho(q)$ had no interior maximum, and $M(q)$ agreed with $\mathcal{T}$ from exact enumeration to 2×10^{-14}.

Theorem 3. *Let $q \in (0, 1]$, $d \geq 1$, $B \geq 1$ and $D = d + 2 - q$. For $\beta \in [0, 1)$, the unique root in $(1, 1/\beta)$ ($1/\beta = +\infty$ at $\beta = 0$) and smallest root in $(1, \infty)$ of $D(z-1)(1-\beta^2 z) = Bq(1-\beta^2 z^2)$ is*

$$z_\beta(q, B) = \frac{2(D + Bq)}{D(1+\beta^2) + \sqrt{D^2(1-\beta^2)^2 + 4\beta^2 B^2 q^2}}.$$

It decreases strictly in β from $z_0(q, B) = 1 + Bq/D$ to 1, and $\rho(M(q)) \geq 1/z_\beta(q, B)$ for every $\eta > 0$, with equality if and only if $\eta = \eta_\beta(q, B) \triangleq \frac{B(1-\beta^2 z_\beta^2)}{z_\beta(1+\beta)D}$ ($z_\beta = z_\beta(q, B)$; $\eta_0 = \frac{B}{Bq+D}$). Hence $\rho(M(q)) \geq D/(D + Bq)$ for every (η, β), strictly if $\beta > 0$: momentum does not make a gated feature decay faster per step than the best SGD step, at any batch size.

At leading order this is known: Everett and Paquette [2026] show (Section 5, Corollary H.7 and Lemma H.8) that with η tuned freely momentum does not train a single feature faster than SGD, nor lower the sample cost to leading order along any co-scaling ray with $\gamma > 0$; Lemma H.8 gives the SGD equality case above, and Lemma H.3 caps the rate by $-\ln\beta$ and, at small feedback, by about twice the SGD rate. Theorem 3 is the exact statement for every d, B, η and β. For a specific input law, Kidambi et al. [2018] show that stochastic heavy ball does not beat the SGD rate; for dense Gaussian linear regression, Morwani et al. [2026] prove finite-dimensional, discrete-time lower bounds showing that heavy ball does not improve the compute-efficiency frontier of SGD for any spectrum. Throughout this section $0 < q_{\min} < q_{\max} \le 1$, $\kappa_V = q_{\max}/q_{\min}$, $d \ge 1$, $B \ge 1$ and $B_\times = (d+2)/(q_{\max} - q_{\min})$. With $E_{\mathrm{mom}} = \inf_B B\, S_{\mathrm{mom}}(B)$, where $S_{\mathrm{mom}}(B)$ (resp. $S_\beta(B)$) is the fewest steps per decay time at batch B over stable (η, β) (resp. η at fixed β) [Everett and Paquette, 2026, Proposition H.15], Theorems 2 and 3 give:

Corollary 4. *For every fixed $\beta \in [0,1)$, $E_\beta(B) = B\, S_\beta(B)$ increases strictly in $B \ge 1$, so the fewest samples per decay time and critical batch size of the class $\{\beta\}$ are*

$$E_\beta = E_\beta(1) = \frac{1}{\ln z_\beta(q_{\min}, 1)}, \qquad B_\beta = \frac{E_\beta}{S_\beta} = \frac{2\ln(1/\rho_\beta)}{\ln z_\beta(q_{\min}, 1)},$$

with $S_\beta = 1/(2\ln(1/\rho_\beta))$ of Everett and Paquette [2026, Proposition H.15(iv)] and ρ_β the largest root modulus of $r^2 - (1+\beta)\frac{\kappa_V - 1}{\kappa_V + 1} r + \beta$. Moreover E_β increases strictly in β from $E_0 = E_{\mathrm{SGD}} = 1/\ln\frac{d+2}{d+2-q_{\min}}$ to $+\infty$; hence $E_{\mathrm{mom}} = E_{\mathrm{SGD}}$ and $B_{\mathrm{mom}}/B_{\mathrm{SGD}} = S_{\mathrm{SGD}}/S_{\mathrm{mom}}$.

Everett and Paquette [2026, Proposition H.15(ii)] prove $E_{\mathrm{mom}} \le E_{\mathrm{SGD}}$ and, for $\kappa_V > 16$, $E_{\mathrm{mom}} \ge c_\kappa(d+2-q_{\min})/(8q_{\min})$, where the 8 comes from Lemma H.3(ii) and disappears at leading order along a co-scaling ray with $\gamma > 0$ (Lemma H.13(iii)); in Remark H.17 a two-row search finds no point below E_{SGD}, noting that "the factor 8 in Proposition H.15(ii) is a loss in the proof". As there, E_O counts samples per decay time of the excess loss and does not involve the label noise; with label noise the loss floors differ [Everett and Paquette, 2026, after Proposition H.15], so $E_{\mathrm{mom}} = E_{\mathrm{SGD}}$ alone does not compare the samples needed to reach a given loss. Everett and Paquette [2026, Appendix A.4] write: "The critical batch size at a fixed β is open, since the fewest samples E_β are not determined." Corollary 4 gives E_β for every β, $\kappa_V > 1$, d and frequency profile, and Theorems 2 and 3 settle $B - 1 \le B_\times$:

Corollary 5. *If $B - 1 \le B_\times$, then $S_\beta(B) = 1/\ln z_\beta(q_{\min}, B)$ increases strictly in $\beta \in [0,1)$ from $S_0(B) = S_{\mathrm{SGD}}(B) = 1/\ln\big(1 + Bq_{\min}/(d+2-q_{\min})\big)$ to $+\infty$; hence $S_{\mathrm{mom}}(B) = S_{\mathrm{SGD}}(B)$, and for $B - 1 \le B_\times$ the tuned momentum is $\beta = 0$.*

Everett and Paquette [2026, Proposition H.16(i)] prove $S_{\mathrm{SGD}}(B)$ in this range, the bound $S_{\mathrm{mom}}(B) \ge c_\kappa \frac{(\kappa_V - 1)(d+2-q_{\min})}{8\kappa_V(d+2)} S_{\mathrm{SGD}}(B)$ for $\kappa_V > 16$, and $S_\beta(B) \ge S_{\mathrm{SGD}}(B)(1 + o(1))$ along a co-scaling ray with $\gamma > 0$ for the rarest token; their Remark H.17 reports numerically that the tuned β is zero just below $B - 1 = B_\times$ and positive just above it.

Theorem 6. *Put $\bar q = \frac{q_{\max} + q_{\min}}{2}$, $r_P = \frac{\sqrt{\kappa_V} - 1}{\sqrt{\kappa_V} + 1}$, $\beta_P = r_P^2$ and $S_{\mathrm{mom}} = \frac{1}{\ln(1/\beta_P)}$, the large-batch limit of Everett and Paquette [2026, Proposition H.15(i)].*

(a) Curve at fixed β. *Fix $\beta \in [0,1)$, write $z_2 = z_\beta(q_{\min}, B)$, $\tilde B(z) = B\frac{1-\beta z}{1+\beta z}$, and for $z \in (1, 1/\beta)$ set $u_{\mathrm{bal}}(z) = \frac{1}{1+\beta}\big(\bar q + \frac{d+2-2\bar q}{2\tilde B(z)}\big)$. Then $S_\beta(B) = 1/\ln z_\beta^{\mathrm{voc}}(B)$, where $z_\beta^{\mathrm{voc}}(B) = z_2$ if $\tilde B(z_2) \le B_\times + 1$ (in particular if $B \le \frac{1+\beta}{1-\beta}(B_\times + 1)$), attained at $\eta_\beta(q_{\min}, B)$; otherwise $z_\beta^{\mathrm{voc}}(B)$ is the unique root in $(1, 1/\beta)$ (lying in $(1, z_2)$) of*

$$\frac{(z-1)(1-\beta^2 z)}{z}\, u_{\mathrm{bal}}(z)^2 = q_{\max} q_{\min}\Big(1 - \frac{1}{\tilde B(z)}\Big),$$

attained at $\eta = 1/u_{\mathrm{bal}}(z_\beta^{\mathrm{voc}}(B))$*, where the two extreme rows have the same rate.*

(b) Above $B_\times$. *If* $B - 1 > B_\times$*, put* $u_0 = \bar{q} + \frac{d+2-2\bar{q}}{2B}$ *and* $\zeta_0 = z_0^{\mathrm{voc}}(B) = \big(1 - \frac{q_{\max} q_{\min}(1-1/B)}{u_0^2}\big)^{-1}$*; then*

$$\frac{d}{d\beta} \ln z_\beta^{\mathrm{voc}}(B)\Big|_{\beta=0^+} = \frac{\zeta_0^2 q_{\max} q_{\min} (q_{\max} - q_{\min})^2 \big((B-1)^2 - B_\times^2\big)}{2B^2 u_0^4} > 0,$$

so $S_{\mathrm{mom}}(B) < S_{\mathrm{SGD}}(B) = 1/\ln \zeta_0$ *and the tuned* β *is positive.*

(c) Approach to the large-batch limit. *With* $\bar{D} = 2(d+2) - q_{\max} - q_{\min}$*,* $\nu_P = \frac{q_{\max} - q_{\min}}{2\sqrt{q_{\max} q_{\min}}}$ *and* $\Delta_B = \frac{q_{\max} + q_{\min}}{2(q_{\max} q_{\min})^{3/4}} \sqrt{\bar{D}/B}$*, every* $B \geq 1$ *satisfies* $S_{\mathrm{mom}}(B) > \frac{1}{2\,\mathrm{arsinh}((\nu_P + \Delta_B)^{-1})} > S_{\mathrm{mom}}$*, and as* $B \to \infty$

$$S_{\mathrm{mom}}(B) = S_{\mathrm{mom}} + C_{\mathrm{mom}} B^{-1/2} + O(B^{-1}), \qquad C_{\mathrm{mom}} = S_{\mathrm{mom}}^2 \frac{4(q_{\max} q_{\min})^{1/4} \sqrt{\bar{D}}}{q_{\max} - q_{\min}},$$

attained to this order at $\beta = \beta_P + \sigma B^{-1/2}$*,* $\sigma = \frac{2(\sqrt{q_{\max}} - \sqrt{q_{\min}})(q_{\max} q_{\min})^{1/4} \sqrt{\bar{D}}}{(\sqrt{q_{\max}} + \sqrt{q_{\min}})^3}$*, and* $\eta = 1/u_{\mathrm{bal}}(z)$ *at* $z = z_\beta^{\mathrm{voc}}(B)$*.*

Everett and Paquette [2026, after Proposition H.15] write: "For momentum the proposition does not describe the curve of steps against batch size between its two asymptotes, the batch size at which the curve bends, or how close $S_{\mathrm{mom}}(B)$ comes to S_{mom} at a finite batch with gradient noise." Theorem 6 gives the curve at every fixed β (so $S_{\mathrm{mom}}(B)$ is a one-variable minimum over β), shows that $\beta > 0$ starts to pay at $B - 1 = B_\times$, and gives two-sided asymptotics for $S_{\mathrm{mom}}(B) - S_{\mathrm{mom}}$, though no closed form for a bend point. For a two-row vocabulary with $d = 8$, $q_{\max} = 10/11$, $q_{\min} = 1/11$ ($\kappa_V = 10$, $C_{\mathrm{mom}} = 6.6601$), the exact $\sqrt{B}\,(S_{\mathrm{mom}}(B) - S_{\mathrm{mom}})$ is 7.0126, 6.7686, 6.6941, 6.6708 at $B = 10^4, 10^5, 10^6, 10^7$.

3 High effective rank

Corollary 7. *Let* $\mu^\star = 2(1+\beta)/\eta^\star = \lambda_{\max}(\widetilde{\mathcal{M}})$*,* $1/\eta_{\mathrm{EMA}}^\star = (1-\beta)/\eta^\star$ *and* $r_{\mathrm{eff}} = \mathrm{tr}\, H/\lambda_1$*. Then*

$$\frac{\sum_k \alpha_k \lambda_k^2}{\mathrm{tr}\, H} \leq \mu^\star - p\kappa_\beta \gamma\, \mathrm{tr}\, H \leq \max_k(\alpha_k \lambda_k), \qquad \frac{1}{\eta_{\mathrm{EMA}}^\star} = \frac{\gamma\, \mathrm{tr}\, H}{2B}(1+\varepsilon), \quad |\varepsilon| \leq \frac{\max_k |\alpha_k|}{p\kappa_\beta \gamma\, r_{\mathrm{eff}}}.$$

Since $\alpha_k/(p\kappa_\beta) = (p\kappa_\beta)^{-1} + \gamma(2+\kappa_k) - 1$*,* $|\varepsilon| = O(1/r_{\mathrm{eff}})$ *for bounded* κ_k *and* $(p\kappa_\beta)^{-1}$*: at high effective rank the threshold for the EMA step is* $2B/(\gamma\, \mathrm{tr}\, H)$*, independent of* β *and of the coordinate fourth moments.*

The limit $2B/(\gamma\, \mathrm{tr}\, H)$ is the high-dimensional noise ceiling; for the isotropic gated model ($H = qI_d$, $\gamma = 1/q$) the exact threshold is $1/\eta_{\mathrm{EMA}}^\star = \frac{d+2-q}{2B} + \frac{q}{2}\frac{1-\beta}{1+\beta}$ [Everett and Paquette, 2026, eq. (6)], whose leading term $d/(2B)$ is $\gamma\, \mathrm{tr}\, H/(2B)$, and Theorem 1(a) reduces to it. A test along the top eigenvector alone sees $\mu_{1\mathrm{D}} = \widetilde{\mathcal{M}}_{11} = \lambda_1\big(1 + p\kappa_\beta(\gamma(3+\kappa_1) - 1)\big)$, which at $\gamma = 1$ is the one-dimensional threshold of Andreyev et al. [2026, eq. (10)]; it is necessary for every β since $\mu^\star \geq \widetilde{\mathcal{M}}_{11}$ (at $\beta = 0$, Mulayoff and Michaeli, 2024, Proposition 9), and misses the trace term that dominates at high effective rank. Table 1 shows the gap.

From Gaussian to Rademacher features $\eta^\star$ moves by 1.32% and the one-direction threshold by a factor 10.5; at $\kappa = -2$ Corollary 7 gives a ratio of at least 144.460 (actual 144.464). Theorem 1(c)–(e) was checked against $\mathcal{T}$ built by exact enumeration of discrete feature laws (e.g. predicted prefactor 7.438933 vs. iterated second-moment value 7.438932). Every number is printed by `paper/checks/note-v2-checks.py` (parts A, C, D and H–K) in https://github.com/apis-os/hb-sgd-mean-square.

Table 1: $P = 256$, $\lambda_k = k^{-1/2}$ ($r_{\text{eff}} = 30.57$), $\beta = 0.9$, $B = 4$, $\gamma = 1$, $\kappa_k \equiv \kappa$. The high-dimensional value $2(1-\beta)B/\operatorname{tr} H$ is 0.026169.

κ	-2 (Rademacher)	-1	0 (Gaussian)	2	10
threshold $\eta^\star$	0.026304	0.026133	0.025961	0.025617	0.024223
one-direction threshold	3.8000	0.66087	0.36190	0.19000	0.065517
ratio	144.46	25.29	13.94	7.42	2.70

4 Relation to prior work

For LMS with Gaussian inputs ($B = 1$, $\beta = 0$) the diagonal second moments evolve through a diagonal plus rank-one matrix whose eigenvalues solve a secular equation [Feuer and Weinstein, 1985]; for momentum LMS the characteristic polynomial has a factor of the form $1 - z\widehat{K}(z)$ [Roy and Shynk, 1990]. For Gaussian features, the stationary equation with momentum is due to Liu et al. [2021] and its closed form to Ziyin et al. [2022, Theorem 2], whose denominator is $1 - S(\eta)$ at $\kappa_k \equiv 0$, $\gamma = 1$; Wang et al. [2026, Appendix B] write the loss as a renewal equation with threshold given by its kernel mass; Velikanov et al. [2023] write the loss generating function and, in the divergent regime, the rate from $r\widetilde{U}(r) = 1$ with a prefactor under an exponential ansatz; without momentum the exact loss is in Bordelon and Pehlevan [2022]. For heavy ball with arbitrary non-commuting minibatch Hessians the exact threshold is Cai et al. [2026, Theorem F.1]; Theorem 1(a) is its closed form in this model. For isotropic gated features, Everett and Paquette [2026] give the threshold, an exact renewal equation (their Lemma C.1) and the per-step spectrum (their Lemma H.4). In high dimension the loss obeys a Volterra equation whose Malthusian root gives the rate, with an explicit prefactor [Paquette et al., 2021, Paquette and Paquette, 2021, Lee et al., 2022]. What is added here is the class with coordinate-dependent fourth moments, whose curvature noise is not of the spectrally expressible form used in the generating-function analyses; the finite-P identity $\rho(\mathcal{T}) = 1/z_\star$ at every η; the prefactor on the stable side and the slope of critical slowing down, which we did not find in prior work also for Gaussian features; and the explicit finite-r_{eff} bounds of Corollary 7. A longer note in the same repository treats momentum and batch in EMA form and the rate at a fixed loss floor.

Use of AI. AI assistants were used for translation, LaTeX, numerical checks and review of the proofs; the author checked every statement and number.

A Proofs

Bounded second moments. Write $s_{t+1} = F_t s_t - \eta E \xi_t$ with $E = \binom{I}{0}$, $m_t = \mathbb{E} s_t$, $Q_t = \mathbb{E} s_t s_t^\top$ and $G = \operatorname{blkdiag}(\eta^2 \Sigma_\xi, 0)$. As s_t is independent of (F_t, ξ_t), $m_{t+1} = \bar{F} m_t$ and $Q_{t+1} = \mathcal{T}(Q_t) + G + C_t$ with $C_t = -\eta\big(L(m_t)E^\top + EL(m_t)^\top\big)$, $L(m) = \mathbb{E}[F_t m \xi_t^\top]$ linear in m. Since $Q \mapsto \mathcal{T}(Q) - \bar{F} Q \bar{F}^\top = \mathbb{E}[(F_t - \bar{F}) Q (F_t - \bar{F})^\top]$ preserves the PSD cone, $\rho(\bar{F})^2 \le \rho(\mathcal{T})$; so if $\rho(\mathcal{T}) < 1$, m_t and C_t decay geometrically and $Q_t = \mathcal{T}^t(Q_0) + \sum_{s<t} \mathcal{T}^s(G + C_{t-1-s})$ stays bounded. Conversely, from $w_0 = w_{-1} = 0$, $m_t = 0$ and $Q_t = \sum_{s<t} \mathcal{T}^s(G)$. The adjoint $\mathcal{T}^*$ preserves the PSD cone, so it has an eigenvector $Y \succeq 0$, $Y \neq 0$, with $\mathcal{T}^*(Y) = \rho(\mathcal{T}) Y$, and $\langle Y, Q_t \rangle = \langle Y, G \rangle \sum_{s<t} \rho(\mathcal{T})^s$. For $(u, v) \neq 0$, $(u,v)^\top (G + F_t G F_t^\top)(u,v) = \eta^2 \big(u^\top \Sigma_\xi u + (A_t^\top u + v)^\top \Sigma_\xi (A_t^\top u + v)\big) > 0$ with $A_t = (1+\beta) I - \eta H_{B_t}$, so $G + \mathcal{T}(G) \succ 0$ and $(1 + \rho(\mathcal{T}))\langle Y, G \rangle = \langle Y, G + \mathcal{T}(G) \rangle > 0$: Q_t is unbounded when $\rho(\mathcal{T}) \ge 1$.

Lemma 8. *Let $\mathcal{V}(X) = \mathbb{E}[(H_{B_t} - H) X (H_{B_t} - H)]$. In the eigenbasis of H, for symmetric X, $\mathcal{V}(X)_{kk} = p\gamma\lambda_k \operatorname{tr}(HX) + c_k X_{kk}$ and $\mathcal{V}(X)_{jk} = p(2\gamma - 1)\lambda_j \lambda_k X_{jk}$ ($j \neq k$). $\mathcal{T}$ preserves both the space $\mathbb{S}_{\text{diag}}$ of symmetric $2P \times 2P$ matrices with diagonal $P \times P$ blocks and its orthogonal complement $\mathbb{S}_{\text{diag}}^\perp$ (blocks with zero diagonal), and $\rho(\mathcal{T}) = \rho(\mathcal{T}|_{\mathbb{S}_{\text{diag}}})$ for every $\eta > 0$.*

Proof. $\mathcal{V}(X) = p(\mathbb{E}[xx^\top Xxx^\top] - HXH)$ and $\mathbb{E}[x_j x_l x_m x_k] = \gamma\sqrt{\lambda_j\lambda_l\lambda_m\lambda_k}\,\mathbb{E}[\omega_j\omega_l\omega_m\omega_k]$ vanishes unless every index appears an even number of times. For $j = k$, $l = m \neq k$ gives $\gamma\lambda_k\lambda_l X_{ll}$ and $l = m = k$ gives $\gamma(3{+}\kappa_k)\lambda_k^2 X_{kk}$; for $j \neq k$, $(l,m) = (j,k), (k,j)$ give $2\gamma\lambda_j\lambda_k X_{jk}$ for symmetric X; subtracting $\lambda_j\lambda_k X_{jk}$ gives the formulas, and since $\mathbb{E}F_t$ has diagonal blocks and $\operatorname{tr}(HX) = 0$ when $\operatorname{diag}(X) = 0$, $\mathcal{T}$ preserves both $\mathbb{S}_{\text{diag}}$ and $\mathbb{S}_{\text{diag}}^{\perp}$. $\mathcal{T}^\tau$ is a sum of congruences, hence positive, so $\|\mathcal{T}^\tau\| = \|\mathcal{T}^\tau(I)\|$ in the norm induced by the spectral norm (Russo–Dye); as $I \in \mathbb{S}_{\text{diag}}$, $\|\mathcal{T}^\tau\| = \|(\mathcal{T}|_{\mathbb{S}_{\text{diag}}})^\tau(I)\| \leq \|(\mathcal{T}|_{\mathbb{S}_{\text{diag}}})^\tau\| \leq \|\mathcal{T}^\tau\|$, and Gelfand's formula gives $\rho(\mathcal{T}) = \rho(\mathcal{T}|_{\mathbb{S}_{\text{diag}}})$. □

In coordinates $(\mathbb{E}w_k^2, \mathbb{E}w_k w_k', \mathbb{E}w_k'^2)_k$ ($w' = w_{t-1}$), $\mathcal{T}|_{\mathbb{S}_{\text{diag}}} = \mathcal{A} + \eta^2 p\gamma\, rr^\top$ with $r = (\lambda_k e_1)_k$, $\mathcal{A} = \operatorname{blkdiag}(\widetilde{B}_k)$ and $\widetilde{B}_k = \begin{pmatrix} a_k^2+\eta^2 c_k & -2a_k\beta & \beta^2 \\ a_k & -\beta & 0 \\ 1 & 0 & 0 \end{pmatrix}$. The mean recursion of mode k from $(1,0)$ is χ_k, so $e_1^\top(\widetilde{B}_k - \eta^2 c_k e_1 e_1^\top)^\tau e_1 = \chi_k(\tau)^2$ and, expanding the rank-one perturbation, $e_1^\top \widetilde{B}_k^\tau e_1 = h_k(\tau)/\eta^2$. For $1 + \beta z \neq 0$, solving rows $3, 2, 1$ of $(I - z\widetilde{B}_k)X_k = \nu e_1$ gives $X_{k,3} = zX_{k,1}$, $X_{k,2} = \frac{a_k z}{1+\beta z}X_{k,1}$ and $\frac{d_k(z)}{1+\beta z}X_{k,1} = \nu$, where $d_k(z) = \det(I - z\widetilde{B}_k) = (1-\beta z)\big((1+\beta z)^2 - z a_k^2\big) - \eta^2 c_k z(1+\beta z)$. Setting $\nu = \eta^2$ gives $\widehat{\psi}_k$ (at $c_k = 0$) and $\widehat{h}_k = \eta^2(1+\beta z)/d_k = \widehat{\psi}_k/(1 - c_k z\widehat{\psi}_k)$. With $\delta_k = \frac{\eta\lambda_k}{2(1+\beta)}$, $(1+\beta)^2 - a_k^2 = 2(1+\beta)\eta\lambda_k(1-\delta_k)$ gives $d_k(1) = 2(1-\beta^2)\eta\lambda_k(1-\alpha_k\delta_k)$, so $\widehat{h}_k(1) = \frac{\eta}{2(1-\beta)\lambda_k(1-\alpha_k\delta_k)}$ and $\widehat{K}(1) = S(\eta)$ when no $\alpha_k\delta_k = 1$.

Theorem 1(a). Since $\gamma \geq 1$ and $3{+}\kappa_k \geq 1$, $\widetilde{\mathcal{M}}_{kk} = \lambda_k\big(1{+}p\kappa_\beta(\gamma(3{+}\kappa_k){-}1)\big) \geq \lambda_k > 0$ and $\widetilde{\mathcal{M}}_{jk} > 0$ ($j \neq k$). For the Perron vector $v > 0$ of $\widetilde{\mathcal{M}}$ at $\mu^\star = \lambda_{\max}(\widetilde{\mathcal{M}}) \geq \widetilde{\mathcal{M}}_{11} \geq \lambda_1 > 0$, $(\mu^\star - \alpha_k\lambda_k)v_k = p\kappa_\beta\gamma\sqrt{\lambda_k}(\sqrt{\lambda}^\top v) > 0$ gives $\mu^\star > \max(0, \max_k \alpha_k\lambda_k)$ and $S(\eta^\star) = 1$ at $\eta^\star = 2(1+\beta)/\mu^\star$, the unique root where $\max_k(\alpha_k\delta_k) < 1$ since $\frac{\eta\lambda_k}{1-\alpha_k\delta_k} = \frac{\lambda_k}{\eta^{-1} - \alpha_k\lambda_k/(2(1+\beta))}$ gives $S'(\eta) > 0$ there. Let $G_{\text{diag}} = \operatorname{blkdiag}(\eta^2\operatorname{diag}(\sigma_k^2), 0)$ be the $\mathbb{S}_{\text{diag}}$-projection of $G = \operatorname{blkdiag}(\eta^2\Sigma_\xi, 0)$. By the $z = 1$ block formulas, $Q = (X_k)_k \in \mathbb{S}_{\text{diag}}$ solves $Q = \mathcal{T}(Q) + G_{\text{diag}}$ [cf. Liu et al., 2021] if and only if $X_{k,3} = X_{k,1}$, $X_{k,2} = (1{-}2\delta_k)X_{k,1}$ and $(I{-}N)y = b$, where $y_k = \frac{2\lambda_k X_{k,1}}{\eta(1+\beta)}$, $b_k = \frac{\sigma_k^2}{1-\beta^2} > 0$ and $N = \frac{\eta}{2(1+\beta)}\big(\operatorname{diag}(\alpha_k\lambda_k) + p\kappa_\beta\gamma\lambda\mathbf{1}^\top\big) > 0$ (similar to $\frac{\eta}{2(1+\beta)}\widetilde{\mathcal{M}}$). If $\rho(\mathcal{T}) < 1$, $Q = \sum_{\tau\geq 0}(\mathcal{T}|_{\mathbb{S}_{\text{diag}}})^\tau(G_{\text{diag}}) \succeq 0$ gives $y \geq b > 0$, $Ny < y$ and $\rho(N) < 1$ (Collatz–Wielandt), i.e. $\eta < \eta^\star$. Conversely, for $\eta < \eta^\star$, $\rho(N) < 1$, $y = (I - N)^{-1}b > 0$ and $|1 - 2\delta_k| < 1$ (as $\mu^\star \geq \lambda_1$), so $Q \succ 0$; for small $\varepsilon > 0$, $Q_\varepsilon = Q - \varepsilon G_{\text{diag}} \succ 0$ satisfies $Q_\varepsilon - \mathcal{T}(Q_\varepsilon) = (1-\varepsilon)G_{\text{diag}} + \varepsilon\mathcal{T}(G_{\text{diag}}) \succ 0$ (as $\mathcal{T}(G_{\text{diag}})_{k,3} = \eta^2\sigma_k^2 > 0$), forcing $\rho(\mathcal{T}) < 1$, $\mathbb{E}s_t \to 0$, and $\ell_t \to \ell_\infty$ via Lemma 8 and $\lambda_k X_{k,1} = \lambda_k\widehat{h}_k(1)(\sigma_k^2 + p\gamma\lambda_k\ell_\infty)$.

(b). From $Q_0 = 0$, with $\mathcal{T}_0(Q) = \bar{F}Q\bar{F}^\top$, $\bar{F} = \mathbb{E}F_t$, $Q_t = \sum_{s<t}\mathcal{T}_0^{t-1-s}(\mathcal{T}(Q_s) - \mathcal{T}_0(Q_s) + G)$ and Lemma 8 give $\mathbb{E}w_{t,k}^2 = \sum_{s<t}\psi_k(t-1-s)(\sigma_k^2 + p\gamma\lambda_k\ell_s + c_k\mathbb{E}w_{s,k}^2)$, solved by $\mathbb{E}w_{t,k}^2 = \sum_{s<t}h_k(t-1-s)(\sigma_k^2 + p\gamma\lambda_k\ell_s)$; multiply by λ_k and sum.

(c). $\det(I - z\mathcal{T}|_{\mathbb{S}_{\text{diag}}}) = \big(\prod_k d_k(z)\big)(1 - z\widehat{K}(z))$ since $\eta^2 p\gamma\, r^\top(I - z\mathcal{A})^{-1}r = \widehat{K}(z)$. As $\mathcal{T}|_{\mathbb{S}_{\text{diag}}}$ preserves the PSD cone $\{X : X_{k,1}, X_{k,3} \geq 0,\ X_{k,1}X_{k,3} \geq X_{k,2}^2\}$, $\rho = \rho(\mathcal{T}) > 0$ has an eigenvector $X \neq 0$ in it; with $z = 1/\rho > 0$, $(I - z\widetilde{B}_k)X_k = z\eta^2 p\gamma(r^\top X)\lambda_k e_1$. By the block formulas above, $r^\top X = \sum_k \lambda_k X_{k,1} = 0$ would force every $X_k = 0$; thus $r^\top X > 0$, $d_k(z) > 0$, $X_{k,1} = zp\gamma(r^\top X)\lambda_k\widehat{h}_k(z)$ and $z\widehat{K}(z) = 1$. Any positive root z' where $\widehat{K}$ is finite is a zero of the determinant, so $1/z' \leq \rho$. Finally, $\rho(\mathcal{T}) \geq \rho(\bar{F})^2 \geq \beta$, while at $\beta = 0$ the closed $(\mathbb{E}w_k^2)_k$ block of $\mathcal{T}|_{\mathbb{S}_{\text{diag}}}$ is symmetric with (j,k) entry $\eta^2 p\gamma\lambda_j\lambda_k > 0$ ($j \neq k$) and $(1,1)$ entry $(1 - \eta\lambda_1)^2 + \eta^2 p(\gamma(3+\kappa_1) - 1)\lambda_1^2 \geq 0$.

(d). With $c_k \geq 0$, $h_k \geq 0$, $K_\tau \geq 0$ and $K_0 > 0$. By Pringsheim's theorem on the rational $\widehat{h}_k$, the radius of convergence R_K of $\widehat{K}$ is either $+\infty$ or a positive pole, so $z\widehat{K}(z) \geq K_0 z$ increases strictly from 0 to $+\infty$ on $[0, R_K)$, crossing 1 once at $z_\star = 1/\rho > 1$. For $|z| \leq z_\star$, $z \neq z_\star$, $\operatorname{Re}(z\widehat{K}(z)) = \sum_\tau K_\tau \operatorname{Re} z^{\tau+1} < 1$ (strict at $\tau = 0$); $\frac{d}{dz}(z\widehat{K})(z_\star) = \sum_\tau(\tau+1)K_\tau z_\star^\tau > 0$; and $\widehat{\phi}$ is analytic on $|z| < R_K$. Thus $\widehat{\ell} = z\widehat{\phi}/((1-z)(1-z\widehat{K}))$ has in $|z| \leq z_\star$ only simple poles at 1 and $z_\star$, giving the expansion and $C > 0$.

(e). At $\eta^\star$, $\rho = 1$, $z_\star = 1$, and $c_k \geq 0$ gives $\alpha_k \geq 1$, so $\alpha_k\delta_k < 1$ forces $\delta_k < 1$: the poles of $\widehat{\psi}_k$ lie outside the unit disk and $1 - c_k\widehat{\psi}_k(1) = \frac{1-\alpha_k\delta_k}{1-\delta_k} > 0$, whence $R_K > 1$. Thus $\Phi(z,\eta) = z\widehat{K}(z;\eta)$ is analytic near $(1,\eta^\star)$ with $\partial_z\Phi = \sum_\tau(\tau+1)K_\tau > 0$ and $\partial_\eta\Phi = S'(\eta^\star) > 0$ by (a); the implicit function theorem gives $\rho'(\eta^\star) = -z_\star'(\eta^\star) = S'/\partial_z\Phi$.

Lemma 9. *Let $q \in (0,1]$, $d \ge 1$, $B \ge 1$, $D_q = d+2-q$, $u = 1/\eta > 0$ and $\beta \in [0,1)$ ($1/\beta = +\infty$ at $\beta = 0$). For $z \in (0, 1/\beta)$ set $g_q(u,z) = \Phi_q(u,z) - R_q(z)$, with $\Phi_q(u,z) = (1+\beta z)^2 u^2 - z\big((1+\beta)u - q\big)^2$ and $R_q(z) = \frac{qD_q z(1+\beta z)}{B(1-\beta z)}$.*

(i) $\rho(M(q)) > \beta$, and for $z \in (0,1/\beta)$, $\rho(M(q)) \le 1/z \iff g_q(u,z) \ge 0$ and $\rho(M(q)) < 1/z \iff g_q(u,z) > 0$. Moreover, with $\tilde\eta = \frac{\eta}{1+\beta}$, $\tilde B(z) = B\frac{1-\beta z}{1+\beta z}$, $Z(z) = \frac{z(1+\beta)^2}{(1+\beta z)^2}$ and $\rho_0(q;\tilde\eta,\tilde B) = (1-\tilde\eta q)^2 + \tilde\eta^2 \frac{qD_q}{\tilde B}$, $g_q(u,z) = z(1+\beta)^2u^2\big(\frac{1}{Z(z)} - \rho_0(q;\tilde\eta,\tilde B(z))\big)$.

(ii) Let $0 < q_2 < q_1 \le 1$, $D_i = d+2-q_i$, $B_\times = \frac{d+2}{q_1-q_2}$, $\nu_P = \frac{q_1-q_2}{2\sqrt{q_1q_2}}$ and $A = \frac{(q_1+q_2)(D_1+D_2)}{4q_1q_2}$. For $\tilde B > 0$, $\min_{\tilde\eta>0}\max_{i=1,2}\rho_0(q_i;\tilde\eta,\tilde B) = \frac{W(\tilde B)}{W(\tilde B)+\tilde B}$, where $W(\tilde B) = \frac{D_2}{q_2}$ on $(0, B_\times+1]$ (uniquely at $\tilde\eta = \frac{\tilde B}{\tilde B q_2 + D_2}$) and $W(\tilde B) = \frac{D_2}{q_2} + \nu_P^2\frac{(\tilde B - 1 - B_\times)^2}{\tilde B - 1} = \tilde B\big(\nu_P^2 + \frac{A}{\tilde B} + \frac{\nu_P^2 B_\times^2}{\tilde B(\tilde B-1)}\big)$ on $(B_\times+1,\infty)$ (uniquely at $\tilde\eta = 1/v_{\rm bal}(\tilde B)$, $v_{\rm bal}(\tilde B) = \frac{q_1+q_2}{2} + \frac{d+2-q_1-q_2}{2\tilde B}$, where both $\rho_0(q_i) = 1 - \frac{q_1q_2(1-1/\tilde B)}{v_{\rm bal}(\tilde B)^2}$); $W(\tilde B)$ is continuous and nondecreasing, while $W(\tilde B)/\tilde B$ decreases strictly on $(0,\infty)$ and exceeds $\nu_P^2 + A/\tilde B$.

Proof. (i) $M(q)$ is $\mathcal{T}|_{\mathbb{S}_{\rm diag}}$ at $P = 1$, $\lambda_1 = q$, $\gamma = D_q/q \ge 2$ and $\kappa_1 = 1/\gamma - 2$ ($c_1 = 0$, $\gamma(3+\kappa_1) = \gamma + 1 > 1$, $\widehat K = \frac{qD_q}{B}\widehat\psi_q$). By Theorem 1(c) and the proof of (d), $\rho(M(q)) > 0$ and $\frac{qD_q}{B} z\widehat\psi_q(z)$ increases strictly from 0 to $+\infty$ on $[0,\pi_q)$, crossing 1 at $z_\star(q) = 1/\rho(M(q)) < \pi_q \le 1/\beta$, where π_q is the first positive pole of $\widehat\psi_q(z) = \frac{1+\beta z}{(1-\beta z)\Phi_q(u,z)}$ ($+\infty$ if none). On $(0,1/\beta)$, $\Phi_q(u,0) = u^2 > 0$: at $\beta = 0$, $\Phi_q(u,\cdot)$ is affine; at $\beta \in (0,1)$, it is convex with $\Phi_q(u,z) > u^2(1-\beta z)^2 > 0$ if $a^2 < 4\beta$ and $\Phi_q(u,1/\beta) = u^2(4 - a^2/\beta) \le 0$ if $a^2 \ge 4\beta$, so in all cases $z < \pi_q \iff \Phi_q(u,z) > 0$. Thus on $(0,1/\beta)$, $\rho(M(q)) \le 1/z \iff (z < \pi_q$ and $\frac{R_q(z)}{\Phi_q(u,z)} \le 1) \iff g_q(u,z) \ge 0$ (as $g_q \ge 0 \implies \Phi_q \ge R_q > 0$), and likewise for $<, >$; dividing by $z(1+\beta)^2u^2$ gives the last formula.

(ii) $\rho_0(q_1;\tilde\eta,\tilde B) - \rho_0(q_2;\tilde\eta,\tilde B) = 2(q_1-q_2)\tilde\eta\big(\tilde\eta v_{\rm bal}(\tilde B) - 1\big)$. At $\tilde\eta_{\rm bal} = 1/v_{\rm bal}(\tilde B)$, both $\rho_0(q_i;\tilde\eta_{\rm bal},\tilde B)$ equal $\rho_{\rm bal}$, so $\rho_0(q;\tilde\eta_{\rm bal},\tilde B) - \rho_{\rm bal} = \tilde\eta_{\rm bal}^2(1 - 1/\tilde B)(q-q_1)(q-q_2)$, and $q = 0$ gives $\rho_{\rm bal} = 1 - \frac{q_1q_2(1-1/\tilde B)}{v_{\rm bal}(\tilde B)^2}$. The vertex $\tilde\eta_i = (q_i + D_i/\tilde B)^{-1}$ of $\rho_0(q_i;\cdot,\tilde B)$ satisfies $q_1 + D_1/\tilde B - v_{\rm bal}(\tilde B) = \frac{q_1-q_2}{2} + \frac{d+2-q_1+q_2}{2\tilde B} > 0$, so $\max_i \rho_0(q_i)$ equals $\rho_0(q_1)$ and increases strictly on $[\tilde\eta_{\rm bal},\infty)$, while on $(0,\tilde\eta_{\rm bal}]$ it equals $\rho_0(q_2)$, minimized uniquely at $\min(\tilde\eta_2,\tilde\eta_{\rm bal})$. Put $X = q_2\tilde B + D_2$ and $Y = q_1(\tilde B - 1)$: $X + Y = 2\tilde B v_{\rm bal}(\tilde B)$ and $X - Y = (q_1-q_2)(B_\times + 1 - \tilde B)$, so $\tilde\eta_2 \le \tilde\eta_{\rm bal} \iff X \ge Y \iff \tilde B \le B_\times + 1$ (giving $W(\tilde B) = D_2/q_2$) and, for $s = \tilde B - 1 > B_\times$, $\tilde B^2 v_{\rm bal}(\tilde B)^2 - XY = \big(\frac{X-Y}{2}\big)^2$ divides by q_1q_2s to $W(\tilde B) = \frac{D_2}{q_2} + \nu_P^2\frac{(s-B_\times)^2}{s} = \frac{D_2}{q_2} + \nu_P^2(\sqrt{s} - B_\times/\sqrt{s})^2$. With $\frac{D_2}{q_2} - \nu_P^2(1+2B_\times) = A > 0$, $\frac{W(\tilde B)}{\tilde B}$ equals $\frac{A + \nu_P^2(1+2B_\times)}{\tilde B}$ on $(0, B_\times+1]$ and $\nu_P^2 + \frac{A}{\tilde B} + \frac{\nu_P^2 B_\times^2}{\tilde B(\tilde B-1)}$ on $(B_\times+1,\infty)$, both strictly decreasing and $> \nu_P^2 + A/\tilde B$. □

Theorem 2. Let $J_s = \{q \in (0,1] : \rho(q) \le s\}$. By Lemma 9(i), $J_s = \emptyset$ for $s \le \beta$, while for $s \in (\beta,1)$ and $z = 1/s \in (1, 1/\beta)$, $J_s = \{q \in (0,1] : g_q(1/\eta, z) \ge 0\}$. In q, $g_q(u,z)$ has degree ≤ 2 and $g_0(u,z) = -(z-1)(1-\beta^2 z)u^2 < 0$: if its q^2 coefficient is ≤ 0, $\{g_\cdot(u,z) \ge 0\}$ is convex; if > 0, $g_\cdot(u,z)$ has roots of opposite signs, so $\{q > 0 : g_q(u,z) \ge 0\}$ is a half-line. Thus J_s is an interval, and $s = \max\big(\rho(q_{\min}), \rho(q_{\max})\big) < 1$ gives the second claim.

Theorem 3. By Lemma 9(i), $\rho(M(q)) > \beta$, and on $z \in (1, 1/\beta)$ we have $Z(z) - 1 = \frac{(z-1)(1-\beta^2 z)}{(1+\beta z)^2} > 0$ and $\min_{\tilde\eta>0}\rho_0(q;\tilde\eta,\tilde B) = \frac{D}{D + \tilde B q}$ uniquely at $\tilde\eta = \frac{\tilde B}{\tilde B q + D}$. Thus $\rho(M(q)) \le 1/z$ holds for some $\eta > 0$ if and only if $Z(z) - 1 \le \frac{\tilde B(z) q}{D}$, i.e. $(z-1)\big(1 + \frac{\beta^2 z(z-1)}{1-\beta^2 z^2}\big) \le \frac{Bq}{D}$. On $(1,1/\beta)$ this left-hand side increases strictly in z from 0 to $+\infty$ and (for $\beta > 0$) in β, so its unique root $z_\beta = z_\beta(q,B) \in (1,1/\beta)$ decreases strictly in $\beta \in [0,1)$ from $z_0 = 1 + Bq/D$ to 1; as $D(z-1)(1-\beta^2 z) - Bq(1-\beta^2 z^2)$ is < 0 at $z = 1$ and > 0 at $z = 1/\beta$ ($\beta > 0$), z_β is also its smallest root in $(1,\infty)$, found from $(D + Bq)z^{-2} - D(1+\beta^2)z^{-1} + \beta^2(D - Bq) = 0$. Any $\rho(M(q)) < 1/z_\beta$ would give $z' = 1/\rho(M(q)) \in (z_\beta, 1/\beta)$, a contradiction; at z_β, $D + \tilde B(z_\beta)q = DZ(z_\beta)$ gives equality if and only if $\eta = (1+\beta)\frac{\tilde B(z_\beta)}{DZ(z_\beta)} = \eta_\beta(q,B)$.

For $q_1 = q_{\max} > q_2 = q_{\min} > 0$ and $D_i = d + 2 - q_i$, Theorem 2 and Lemma 9 show that on $z \in (1,1/\beta)$, $\max_j \rho(q_j) \le 1/z$ for some $\eta > 0$ if and only if $\frac{W(\tilde B(z))}{\tilde B(z)} \le \frac{1}{Z(z)-1}$, uniquely at the optimal $\tilde\eta$ of Lemma 9(ii). Here $\frac{W(\tilde B(z))}{\tilde B(z)}$ is continuous and nondecreasing ($= W(B)/B > 0$ at $\beta = 0$; $\to +\infty$ as $z \uparrow 1/\beta$ if $\beta > 0$), while

$\frac{1}{Z(z)-1}$ decreases strictly from $+\infty$ to $\frac{4\beta}{(1-\beta)^2}$, giving a unique root $z_\beta^{\mathrm{voc}}(B) \in (1, 1/\beta)$, continuous in $\beta \in [0,1)$ with $z_\beta^{\mathrm{voc}}(B) \le z_\beta(q_2, B) \to 1$ as $\beta \uparrow 1$, $S_\beta(B) = 1/\ln z_\beta^{\mathrm{voc}}(B)$, and $W(\tilde{B}(z)) = \frac{B(1-\beta^2 z^2)}{(z-1)(1-\beta^2 z)}$ at $z = z_\beta^{\mathrm{voc}}(B)$.

Corollary 4. Put $z_B = z_\beta^{\mathrm{voc}}(B)$, $x_B = \tilde{B}(z_B)$ and $r(x) = W(x)/x$, strictly decreasing on $(0,\infty)$ by Lemma 9(ii). (1) z_B increases strictly: for $1 \le B < B'$, the batch B' gives $\tilde{B}(z_B) = \frac{B'}{B} x_B > x_B$ at $z = z_B$, so there $r(\tilde{B}(z)) < r(x_B) = \frac{1}{Z(z_B)-1}$; as $z \mapsto r(\tilde{B}(z))$ is nondecreasing and $z \mapsto \frac{1}{Z(z)-1}$ decreases strictly, the root $z_{B'}$ lies in $(z_B, 1/\beta)$. (2) x_B increases strictly, since $r(x_B) = \frac{1}{Z(z_B)-1}$ decreases strictly by (1). (3) With $Z(z)-1 = \frac{(z-1)(1-\beta^2 z)}{(1+\beta z)^2}$ and $x_B = B\frac{1-\beta z_B}{1+\beta z_B}$, the root equation $r(x_B) = \frac{1}{Z(z_B)-1}$ reads $B = W(x_B)\frac{(z_B-1)(1-\beta^2 z_B)}{1-\beta^2 z_B^2}$, so $E_\beta(B) = \frac{B}{\ln z_B} = W(x_B) f(z_B)$ with $f(z) = \frac{z-1}{\ln z}\big(1 + \frac{\beta^2 z(z-1)}{1-\beta^2 z^2}\big)$. (4) $W \ge D_2/q_2 > 0$ is nondecreasing, so $W(x_B)$ is nondecreasing in B by (2); $f > 0$ increases strictly on $(1, 1/\beta)$, since $\frac{d}{dz}\frac{z-1}{\ln z} = \frac{\ln z - 1 + 1/z}{\ln^2 z} > 0$ and $\frac{z(z-1)}{1-\beta^2 z^2}$ increases, so $f(z_B)$ increases strictly by (1). Hence $E_\beta(B)$ increases strictly in $B \ge 1$. At $B = 1$, $\tilde{B}(z_1) \le 1 < B_\times + 1$ gives $W(\tilde{B}(z_1)) = D_2/q_2$, $z_1 = z_\beta(q_2, 1)$ and $E_\beta = E_\beta(1) = 1/\ln z_1$. By Theorem 3, E_β increases strictly in $\beta \in [0,1)$ from $E_0 = E_{\mathrm{SGD}}$ to $+\infty$, giving $E_{\mathrm{mom}} = E_{\mathrm{SGD}}$ and, via $B_O = E_O/S_O$ [Everett and Paquette, 2026, Proposition H.15], $B_{\mathrm{mom}}/B_{\mathrm{SGD}} = S_{\mathrm{SGD}}/S_{\mathrm{mom}}$.

Corollary 5. If $B-1 \le B_\times$, every $\beta \in [0,1)$ and $z \in (1, 1/\beta)$ satisfy $\tilde{B}(z) \le B \le B_\times + 1$, so $W(\tilde{B}(z)) = D_2/q_2$ and $z_\beta^{\mathrm{voc}}(B) = z_\beta(q_2, B)$ (cf. Everett and Paquette, 2026, Proposition H.16(i), Lemma H.13(v)); Theorem 3 gives strict increase of $S_\beta(B) = 1/\ln z_\beta(q_2, B)$ in β.

Theorem 6. *(a).* At z_2, $\frac{D_2}{\tilde{B}(z_2) q_2} = \frac{1}{Z(z_2)-1}$: if $\tilde{B}(z_2) \le B_\times + 1$ (implied by $B\frac{1-\beta}{1+\beta} \le B_\times + 1$), $\frac{W(\tilde{B}(z_2))}{\tilde{B}(z_2)} = \frac{1}{Z(z_2)-1}$ gives $z_\beta^{\mathrm{voc}}(B) = z_2$ at $\eta_\beta(q_2, B)$; otherwise $\frac{W(\tilde{B}(z_2))}{\tilde{B}(z_2)} > \frac{1}{Z(z_2)-1}$, so $z_\beta^{\mathrm{voc}}(B) \in (1, z_2)$, where $\tilde{B}(z) > B_\times + 1$ and $\frac{q_1 q_2 (1 - 1/\tilde{B}(z))}{v_{\mathrm{bal}}(\tilde{B}(z))^2} = 1 - \frac{1}{Z(z)}$ at $\eta = 1/u_{\mathrm{bal}}(z)$ gives the balanced equation (strictly increasing on the left and nonincreasing on the right on $(1, 1/\beta)$).

(b). For $B - 1 > B_\times$ and small $\beta \ge 0$, $\tilde{B}(z_\beta^{\mathrm{voc}}(B)) > B_\times + 1$. Since β enters $\frac{B(1-\beta^2 z^2)}{(z-1)(1-\beta^2 z)}$ only as β^2 and $\frac{d}{d\beta}\tilde{B}(z)\big|_{0^+} = -2B\zeta_0$, differentiating $W(\tilde{B}(z)) = \frac{B(1-\beta^2 z^2)}{(z-1)(1-\beta^2 z)}$ at $\beta = 0^+$ gives $-2B\zeta_0 W'(B) = -\frac{B\zeta_0}{(\zeta_0-1)^2}\frac{d}{d\beta}\ln z\big|_{0^+}$. With $W'(B) = \nu_P^2\big(1 - \frac{B_\times^2}{(B-1)^2}\big)$ and $\frac{\zeta_0 - 1}{B-1} = \frac{\zeta_0 q_1 q_2}{B u_0^2}$, $\frac{d}{d\beta}\ln z\big|_{0^+} = 2(\zeta_0 - 1)^2 W'(B)$ equals the stated positive slope, so $\beta \mapsto z_\beta^{\mathrm{voc}}(B)$ attains its maximum on $[0,1)$ at some $\beta > 0$.

(c). Let (η, β) be stable with $\max_j \rho(q_j) = 1/z$, $z = e^\Lambda$ $(< 1/\beta)$, and $\theta = \frac{1+\beta z}{1-\beta z} \ge 1$. Lemma 9 gives $\nu_P^2 + \frac{A}{B}\theta < \frac{W(B/\theta)}{B/\theta} \le \frac{1}{Z(z)-1}$, while $\sqrt{Z(z)} = \frac{e^{\Lambda/2}(1+\beta)}{1+\beta z} = \cosh\frac{\Lambda}{2} + \theta^{-1}\sinh\frac{\Lambda}{2}$ gives $Z(z) - 1 > \sinh^2\frac{\Lambda}{2}\big(1 + \frac{2\coth(\Lambda/2)}{\theta}\big)$. By Cauchy–Schwarz, $\big(\nu_P + \sqrt{2\coth(\Lambda/2) A/B}\big)^2 \le (\nu_P^2 + \frac{A}{B}\theta)\big(1 + \frac{2\coth(\Lambda/2)}{\theta}\big) < \sinh^{-2}(\Lambda/2)$, so $\sinh(\Lambda/2) < \nu_P^{-1} = \sinh(\frac{1}{2}\ln\beta_P^{-1})$ [Everett and Paquette, 2026, cf. Lemma H.14], $2\coth(\Lambda/2) > 2\sqrt{1+\nu_P^2} = \frac{q_1+q_2}{\sqrt{q_1 q_2}}$, and $\sinh(\Lambda/2)^{-1} > \nu_P + \Delta_B$ at the attained maximum of $\beta \mapsto z_\beta^{\mathrm{voc}}(B)$, yielding the lower bound and its $B \to \infty$ expansion. Conversely, put $\theta_0 = \frac{q_1 - q_2}{(q_1 q_2)^{1/4}\sqrt{D}}$, $\beta = \beta_P + \sigma B^{-1/2}$ $(\sigma = \frac{2\beta_P}{\theta_0})$, $\phi(\beta) = \frac{4\beta}{(1-\beta)^2}$ and fix $m > 0$. For large B, $\theta = \theta_0\sqrt{B} - m > \frac{1+\beta}{1-\beta}$ defines $z = \frac{\theta-1}{\beta(\theta+1)} = \beta_P^{-1}(1 - \frac{4}{\theta_0}B^{-1/2} + O(B^{-1})) \in (1, 1/\beta)$ with $\tilde{B}(z) = B/\theta > B_\times + 1$ and $Z(z) - 1 = \phi(\beta)^{-1} - \frac{(1+\beta)^2}{4\beta}\theta^{-2}$. Since $\phi(\beta_P) = \nu_P^2$, $\phi'(\beta_P)\sigma = 2\sqrt{1+\nu_P^2}\frac{\nu_P^2}{\theta_0} = A\theta_0$ and $\theta^{-2} = \theta_0^{-2}B^{-1} + O(B^{-3/2})$, Lemma 9(ii) gives $\frac{1}{Z(z)-1} - \frac{W(B/\theta)}{B/\theta} = (Am + c_0)B^{-1} + O(B^{-3/2})$ (c_0 independent of m), which is > 0 for large m and B. Thus $z_2 \ge z_\beta^{\mathrm{voc}}(B) > z$, $S_{\mathrm{mom}}(B) \le S_\beta(B) < 1/\ln z = S_{\mathrm{mom}} + C_{\mathrm{mom}} B^{-1/2} + O(B^{-1})$ ($C_{\mathrm{mom}} = \frac{4S_{\mathrm{mom}}^2}{\theta_0}$), and $\tilde{B}(z_2) = \frac{D_2}{q_2}(Z(z_2) - 1) \to \frac{D_2}{q_2}\nu_P^{-2} > 1 + 2B_\times > B_\times + 1$ puts $S_\beta(B)$ at $\eta = 1/u_{\mathrm{bal}}(z_\beta^{\mathrm{voc}}(B))$.

Corollary 7. The Rayleigh quotient of $\widetilde{\mathcal{M}}$ at $\sqrt{\lambda}/\sqrt{\mathrm{tr}\, H}$ gives the lower bound, and Weyl's inequality $\lambda_{\max}(\widetilde{\mathcal{M}}) \le \max_k(\alpha_k\lambda_k) + p\kappa_\beta\gamma\,\mathrm{tr}\, H$ the upper one. Since $\sum_k \lambda_k^2 \le \lambda_1\,\mathrm{tr}\, H$, both bounds have modulus at most $\max_k|\alpha_k|\lambda_1$, giving $|\varepsilon| \le \max_k|\alpha_k|/(p\kappa_\beta\gamma\, r_{\mathrm{eff}})$ via $1/\eta_{\mathrm{EMA}}^\star = \mu^\star/(2\kappa_\beta)$.

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